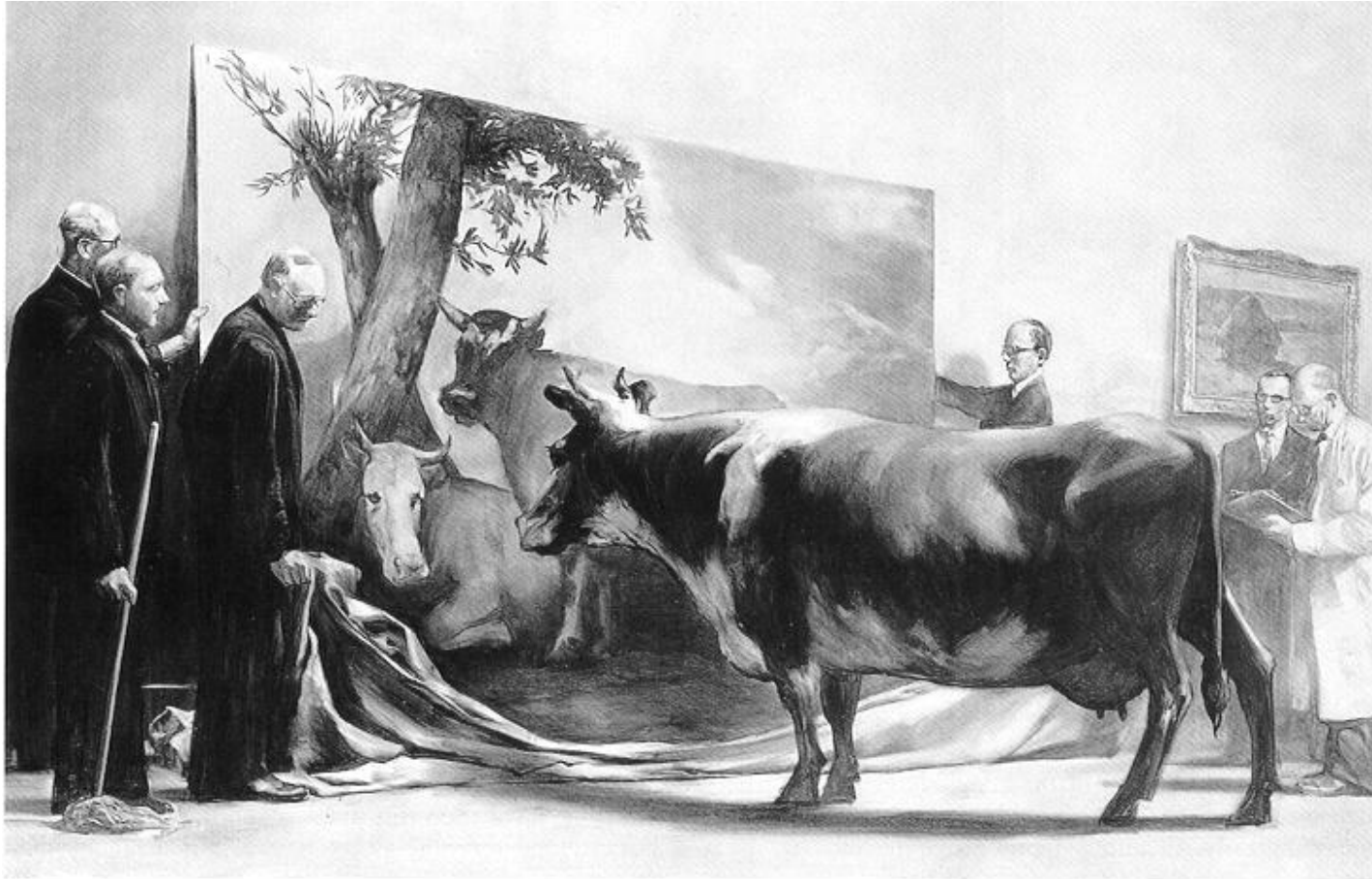


https://www.youtube.com/watch?v=eKLzgMYhGFw&list=RD eKLzgMYhGFw&start_radio=1

ENVX2001 – Applied Statistical Methods



Lecture 1- Multivariate Approaches

Prof Mathew Crowther, SoLES

mathew.crowther@sydney.edu.au

ENVX2001 – Applied Statistical Methods



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ENVX2001–Applied Statistical Methods



Lecture 1- Multivariate Approaches

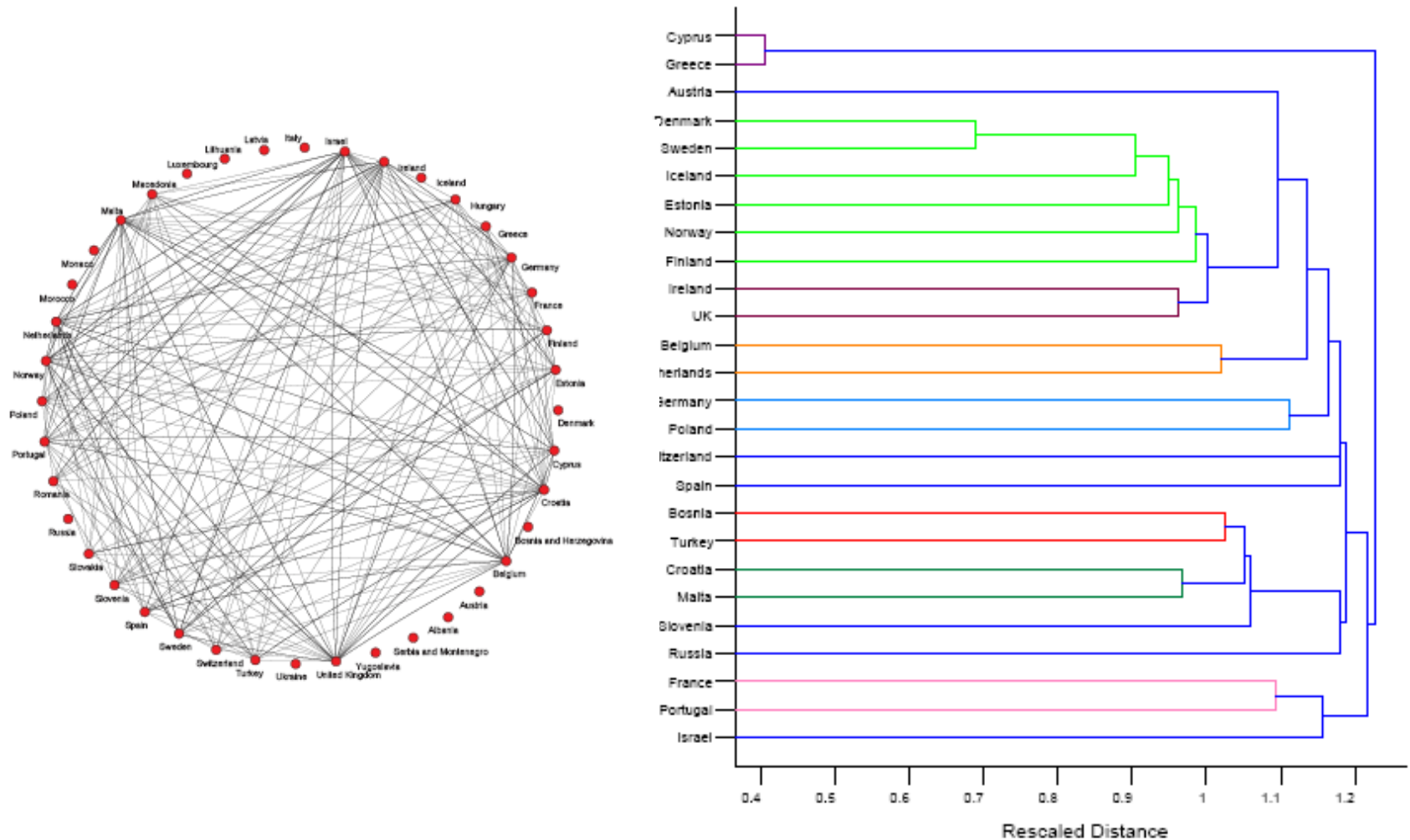
Prof Mathew Crowther, SoLES

mathew.crowther@sydney.edu.au

Scoreboard: Eurovision Song Contest 2008 Final

[illegible]

Fenn et al. 2005. How does Europe Make Its Mind Up? Connections, cliques, and compatibility between countries in the Eurovision Song Contest. *Physica A*, 360: 576-598.



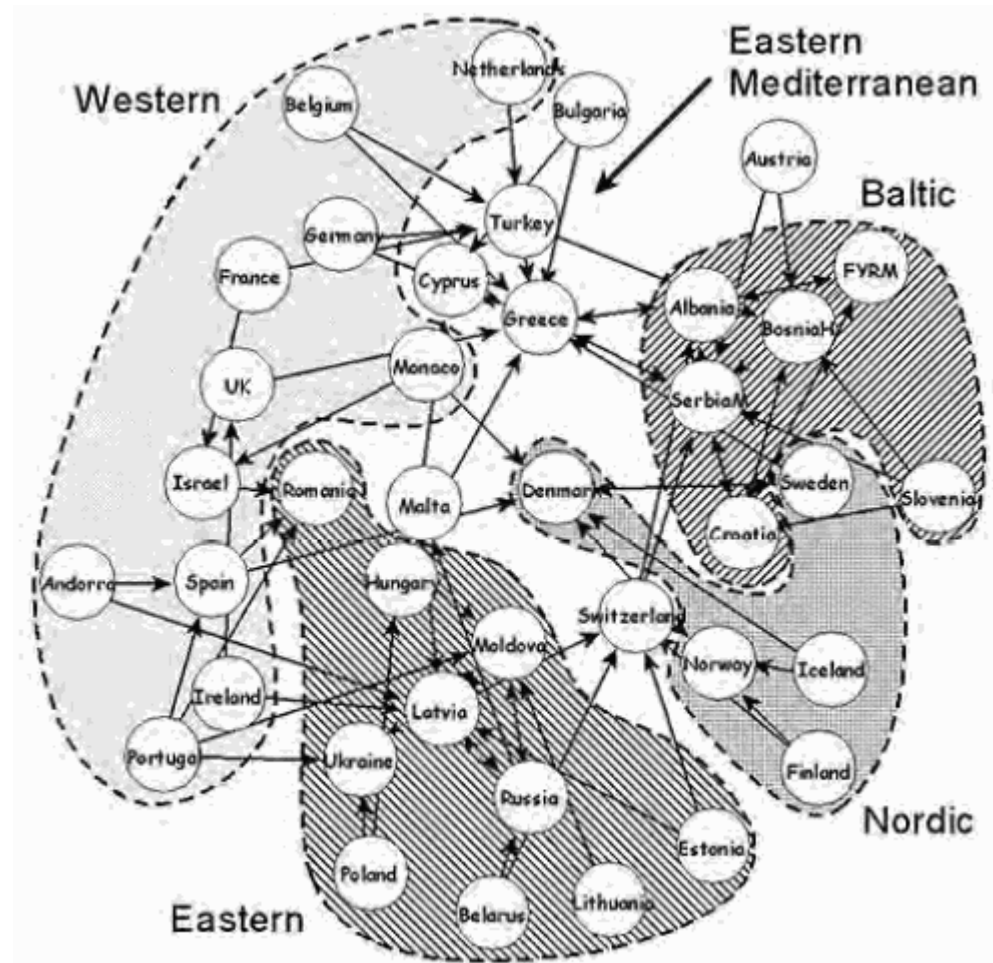


Figure 4. Friendship Blocs in the Eurovision Song Contest

Eurovision: bloc voting or cultural clusters

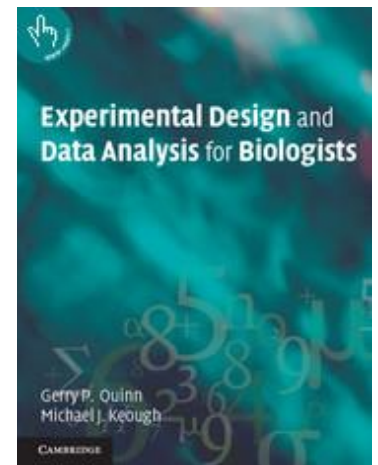
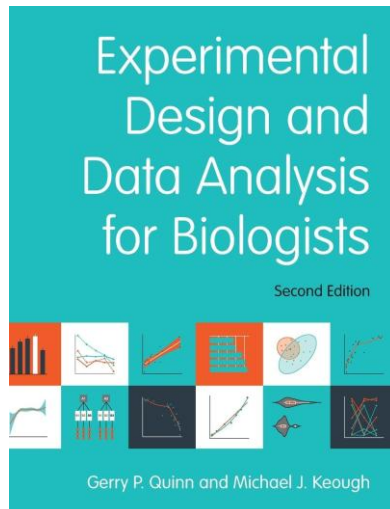
http://andrewgelman.com/2007/05/30/eurovision_and/

Outline

- Multivariate data
- Hypothesis testing and data exploration
- Important techniques including
 - PCA
 - Cluster analysis
 - nMDS
 - Permutational approaches (ANOSIM, PERMANOVA)
 - MANOVA

Readings

- Quinn & Keough (2002) – Chapter 15
- Quinn & Keough (2023) – Chapter 14
- Han, S. Y., Filippi, P., Román Dobarco, M., Harianto, J., Crowther M. S., and Bishop, T. F. A. (2023). Multivariate analysis for soil science. In *'Encyclopedia of Soils in the Environment (Second Edition)'*. (Ed. M. J. Goss and M. Oliver) pp. 499-508. Academic Press: Oxford.





Cyanide and Happiness © Explosm.net

Specific themes

(“What don’t you know?”)

- Ordination
- Principal Components Analysis (PCA)
- Nonmetric multidimensional scaling (nMDS)
- Cluster analysis
- MANOVA / PERMANOVA

Goals for this section

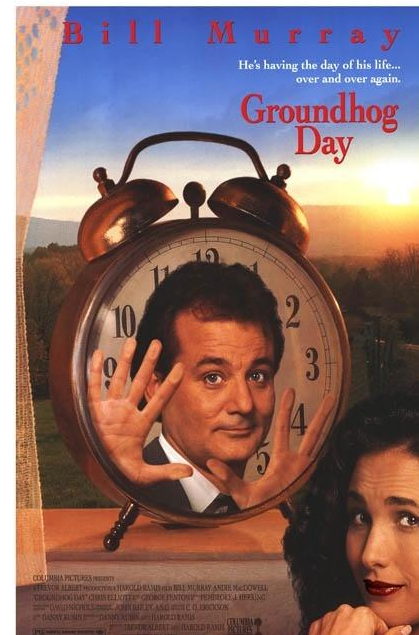
- Use a range of multivariate techniques confidently
- Choose appropriate analyses for different questions
- Analyse your own data effectively
- Interpret and critique others' data
- Run, interpret, and present analyses in R
- Understand statistical outputs from the literature (not all in R)

Multivariate Analysis can be challenging– but it doesn't have to be

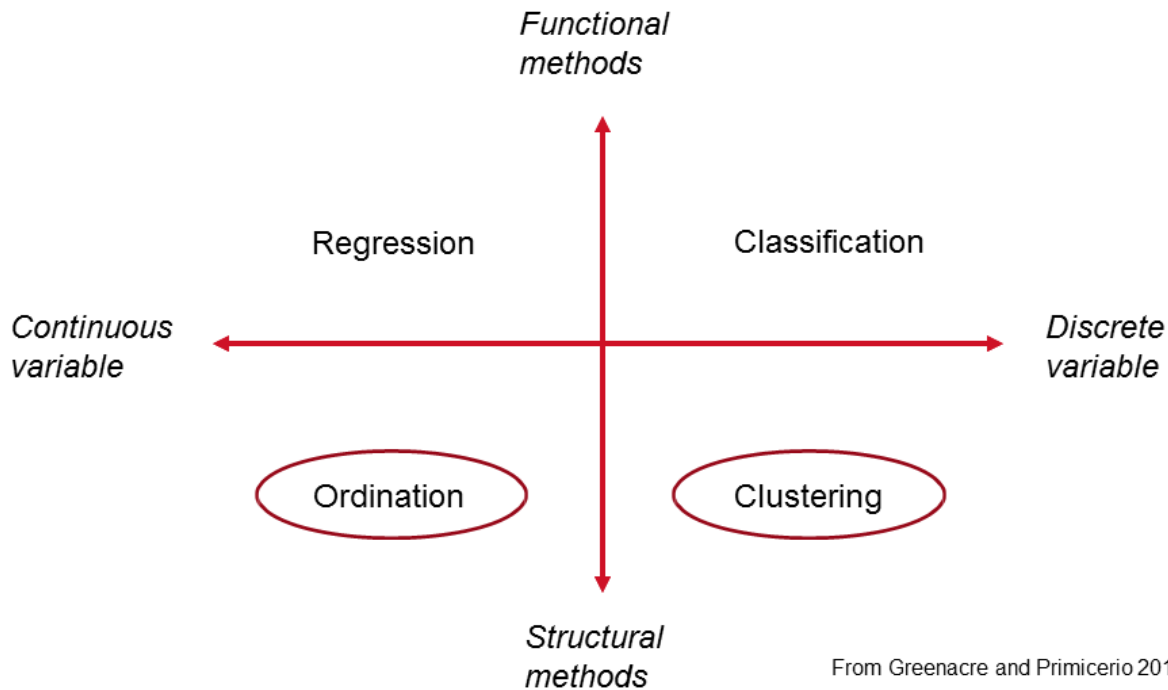
- What do you do if you don't understand something?
 - Ask questions via **Ed Discussion** – I will get to them quickly
 - Email me mathew.crowther@sydney.edu.au
 - Read Quinn & Keough (2002, 2023), it's very good and free online in library!
 - Ask questions in labs to tutors

This lecture

- Multivariate data
 - Multivariate questions
 - Multivariate statistics
- Central issues
 - Linear combinations
 - Distance measures
 - Transformations
 - Standardisation

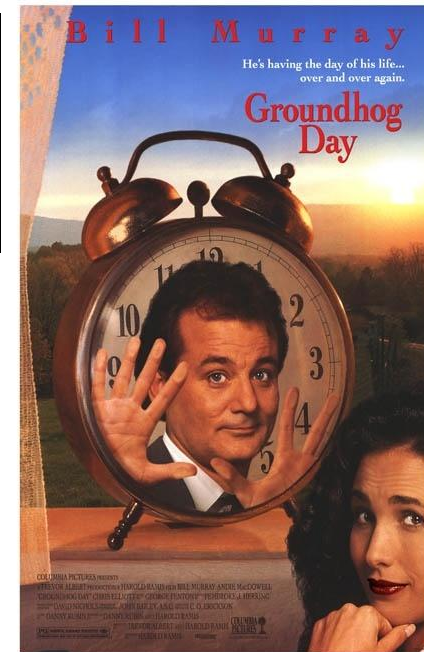


Multivariate Methods



First Steps

- Linear combinations of variables
- Distance (dissimilarity) measures
- ***Theme for the practicals*** - familiarity with both approaches (using PCA, Clustering & nMDS/ANOSIM/PERMANOVA) to address environmental/biological questions in a generalized system



What are multivariate data?

- Multiple **response** variables
- Multivariate data in environmental applications most often occur
 - in the form of counts of species abundances in assemblages, where each species is a variable.
 - when assessing physical properties of environments (e.g. habitat structure, soil, water quality)
- Multivariate data often violate assumptions of normality, independence, and homogeneity of variance.
- Specialised approaches and methods needed

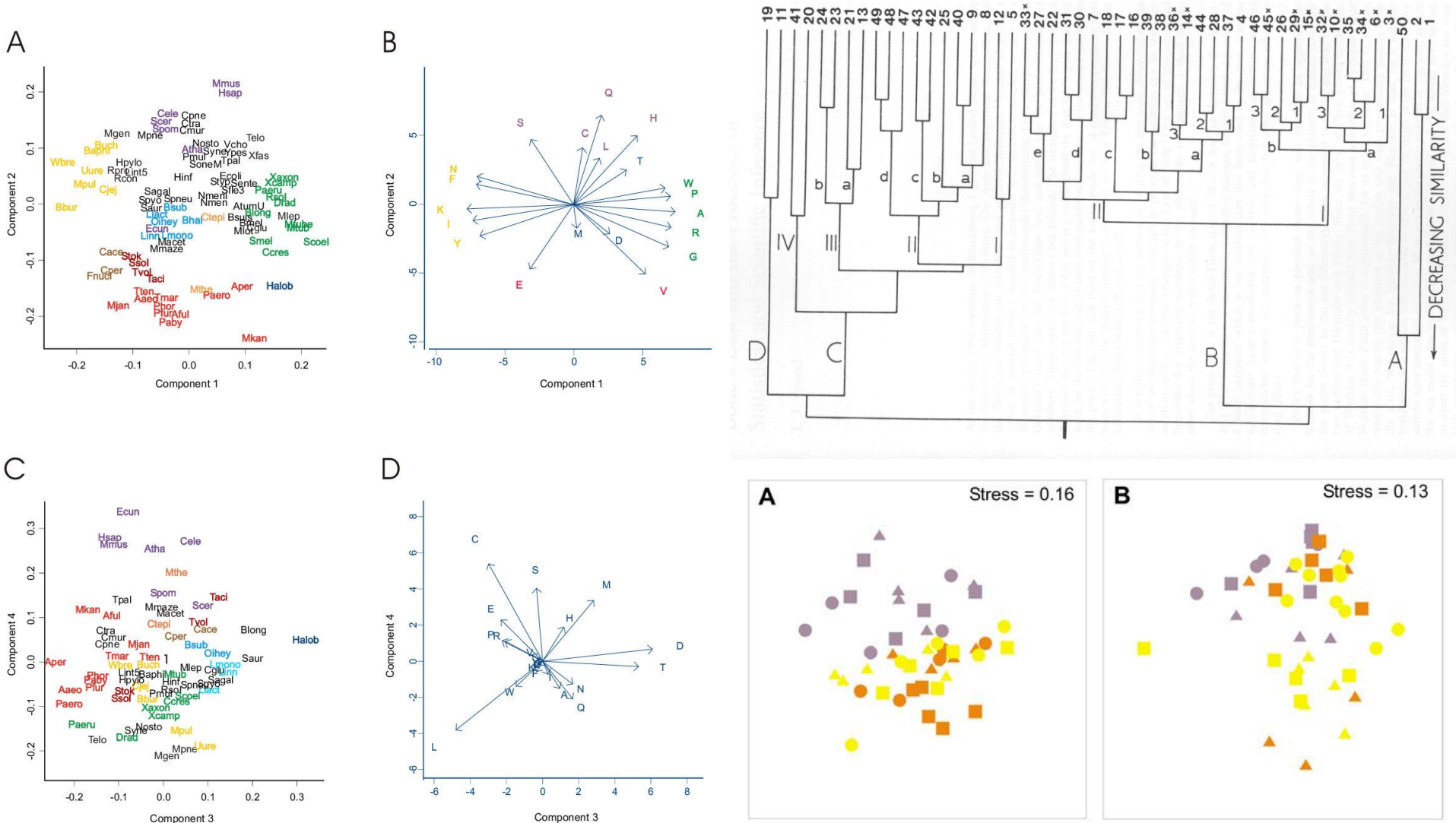
What are multivariate statistics?

- Statistics that allow us to look at how multiple variables change together
 - How do 50 species in a community react to an environmental perturbation?
 - How do 10 measures of habitat quality change in response to disturbance?
- Should we do multiple ANOVAs?
- Multivariate stats allow us to “condense” information for simplicity

Typical multivariate data from biological research examining

- Changes in community composition
 - Species in different sites
- Changes in water chemistry
 - Multiple ions and characteristics
- Changes in soil characteristics
 - Chemicals and spectra
- Changes in habitat characteristics
 - Assessing many landscape features simultaneously
- Changes in organismal characteristics
 - Life history evolution, characterising organism phenotype using multiple traits

PCA, cluster analysis and nMDS



General principles

Data reduction

Assessing multivariate responses



General principles

Data reduction

Assessing multivariate responses

Consider habitat complexity and insect assemblages

- Rumored to be important in driving diversity (species richness, community composition etc.)
- What can we measure?
 - Canopy cover
 - Shrub cover
 - Litter (ground) cover
 - Numbers of stems
 - Diversity (trees, shrubs, understorey)
 - Ground characteristics (logs, rocks, dirt cover)
 - Soil characters (moisture, nutrients)

Variable	Plot 1	Plot 2	Plot 3	Plot 4
Canopy cover (%)	85	65	40	20
Shrub cover (%)	30	45	25	35
Litter (ground) cover (%)	80	60	35	15
Number of stems (per 100 m²)	45	60	30	20
Diversity (trees – spp.)	12	10	7	5
Diversity (shrubs – spp.)	8	12	6	10
Diversity (understorey – spp.)	15	18	10	14
Ground – logs (%)	25	20	10	5
Ground – rocks (%)	5	10	20	15
Ground – bare soil (%)	10	20	40	70
Soil moisture (% volumetric)	35	25	15	10
Soil nutrients (index 0–10)	8.5	7	5.5	4.5

Double-dipping when measuring habitat complexity

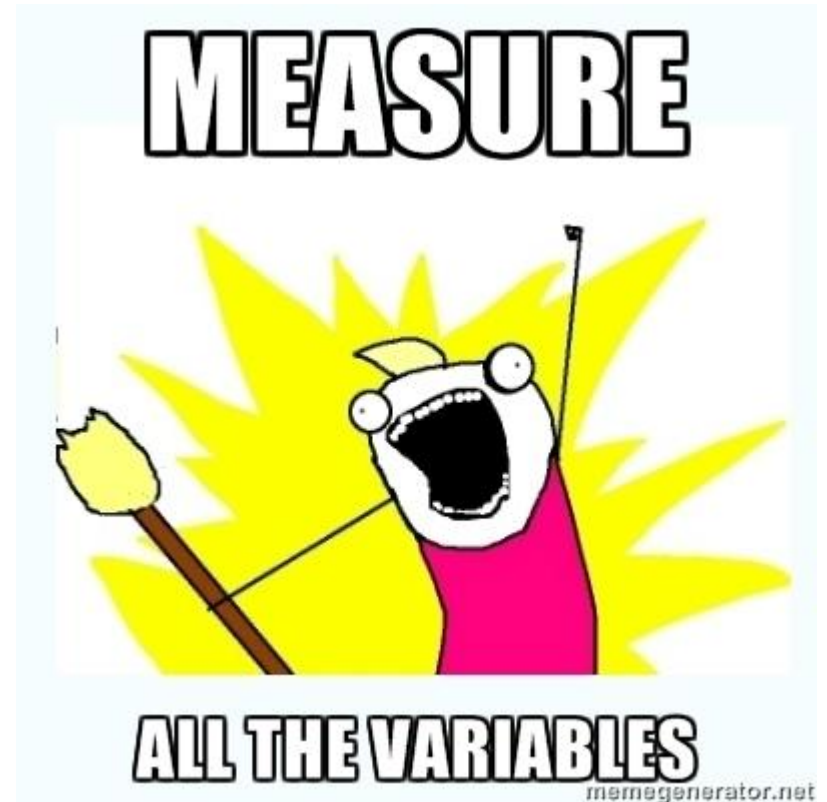
- Associations between measures?
 - Canopy cover
 - Shrub cover
 - Litter (ground) cover
 - Numbers of stems
 - Diversity (trees, shrubs, understorey)
 - Ground characteristics (logs,
 - Soil characters
- Possibility of correlations among factors?

What we do – reducing variables

- Summarise existing data more simply by reducing existing variables to a smaller number of newly derived variables
- Reveal patterns that would not be found by examining each variable separately
- Desirable in many contexts
 - habitat and vegetation assessments
 - organismal traits
 - water quality
 - soil properties
- Principal Components Analysis (PCA) – useful way to generate simple metrics from complex multivariate datasets

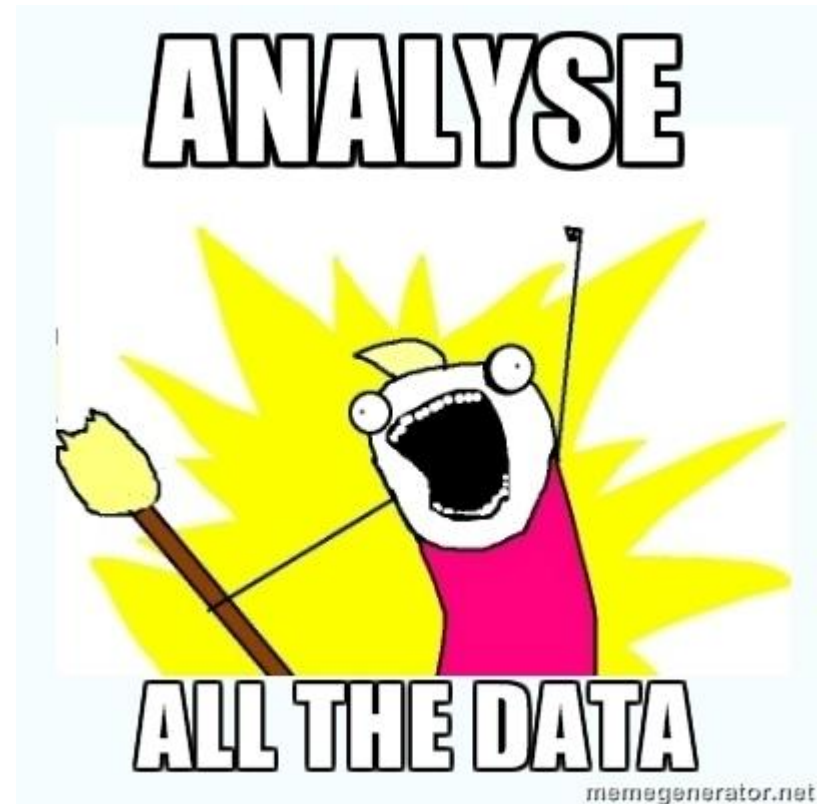
Hypothesis-free junk?

the PCA is often just the start

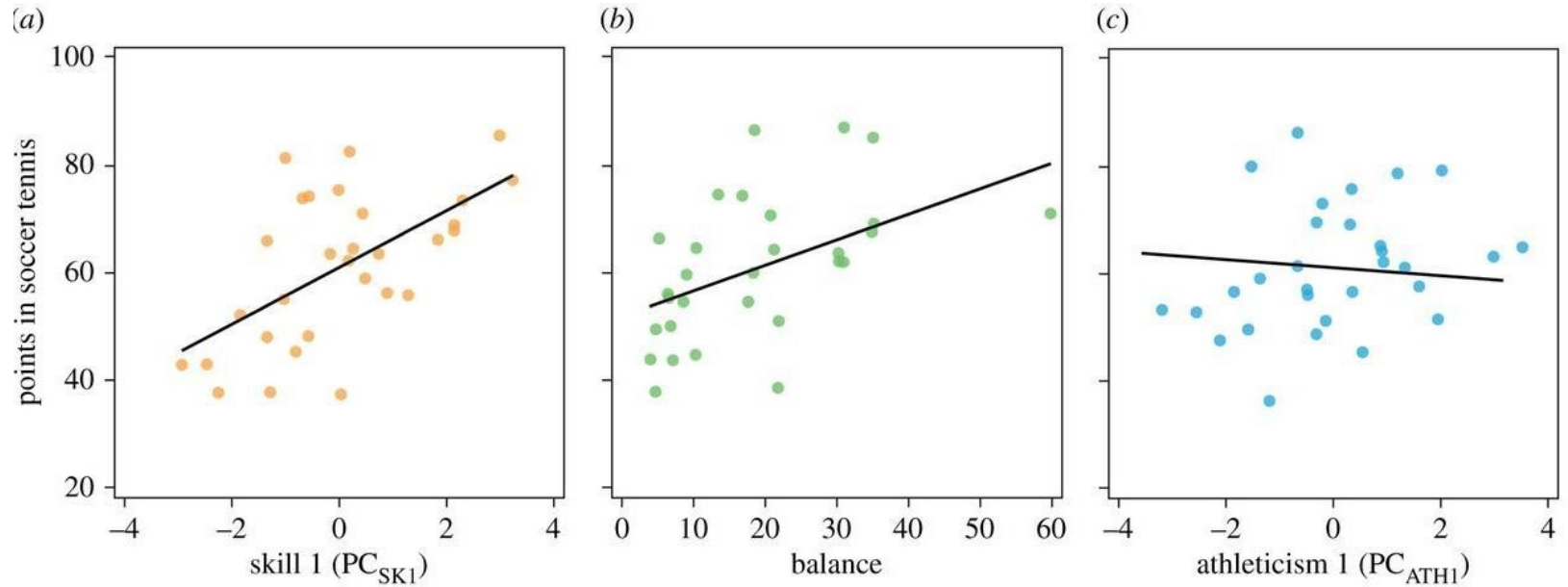


Hypothesis-free junk?

the PCA is often just the start



Football



Wilson *et al.* (2017)

Consider hypotheses examining assemblages from landscapes managed differently

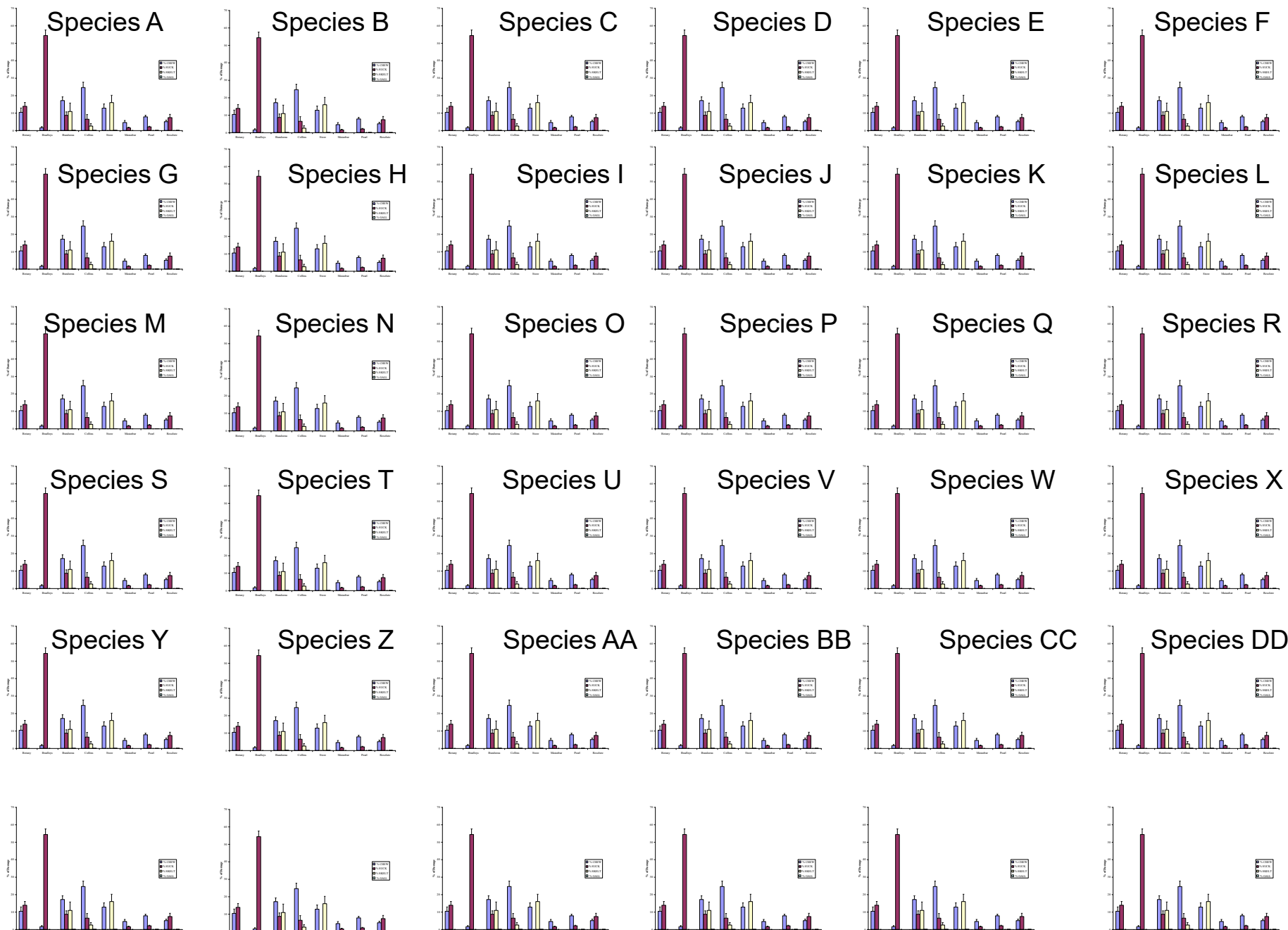
- Different from univariate approaches?
- End up with counts of species at each site.
- What are the hypotheses?





Control Control Disturbed Disturbed

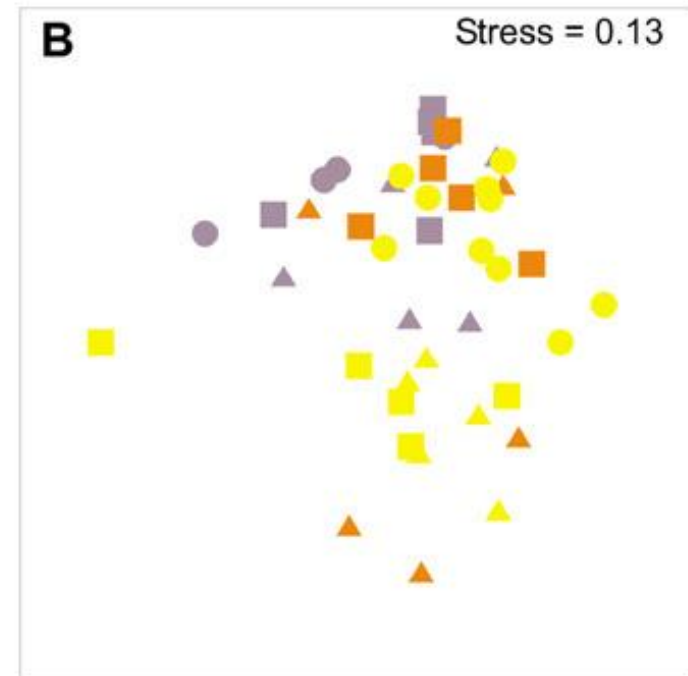
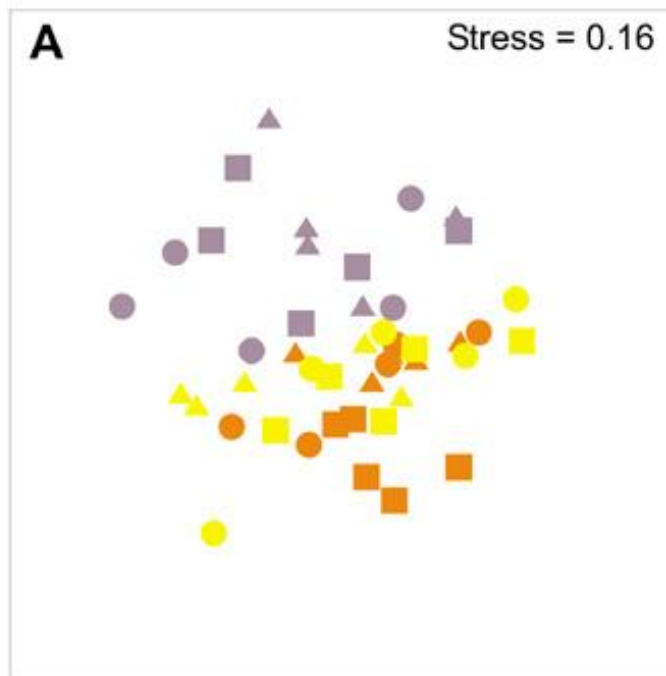
Diptera	Tipulidae	20	58		4
Diptera	Muscidae			1	1
Diptera	Stratiomyidae	29			
Diptera	Empididae	97	18	23	82
Diptera	Athericidae		5		7
Diptera	Thaumaleidae	6		4	5
Diptera	Psychodidae	1	5		1
Diptera	Tanyderidae	178	106	88	102
Diptera	Dolichopodidae		1		
Diptera	Ceratopogonida	249	20	1	5
Diptera	Chironomidae	169	132	117	11
Diptera	?BBBB	1		1	
Diptera	?DDDD			1	
Diptera	?HHHH				1
Diptera	?MMMM				201
Ostracod	Darwinulidae				1
Coleopter	Hydraenidae	6	8		1
Coleopter	Staphylinidae				
Coleopter	Leiodidae	2	30	10	1



General principles

Data reduction

Assessing multivariate responses



What we do – community structure

Three main categories of multivariate analysis that are common in biology and environmental science:

- (i) **clustering** - generally concerned with the identification of clusters or groups of observations (or species). Most often used for the generation of hypotheses in the beginning of a study or for exploratory purposes
- (ii) **ordination** – comprehensible rearrangement of elements. We want to produce an ordination in order to visualize patterns in the multivariate cloud of data (by reducing dimensions in some way)
- (iii) **statistical tests of hypotheses** - rigorously test explicit hypotheses concerning those patterns by reference to *a priori* groups or relationships with predictors, such as environmental variables

Our focus on where we have a specific hypothesis in mind and have either

- sampled natural systems in a structured way or
- done a manipulative experiment

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